Predictions based on Performance Ratings

Elo ratings

Elo rating system

- Is initially developed by Hungarian-american professor of physics **Arpad Elo.**
- Initially was used to rank chess players.
- Can be used to predict the game outcome for any two-players or two-teams game.



Arpad Elo

Elo rating system

- The basic idea
- Assign performance rating to each team/player. This rating is going to show the ability of the team to win.
- After each game update the rating of the player based on the outcome.
- Use the ability scores to predict the match outcome of the game

Elo rating system

The general formula

The expected probability of winning a game by Team A will be calculated by the following formula.

For home team

$$\gamma^H = \frac{1}{1 + c^{(R^A - \theta - R^H)/d}}$$

For away team

$$\gamma^A = 1 - \frac{1}{1 + c^{(R^A - \theta - R^H)/d}}$$

Where R^H, R^A are home and away teams Ratings, θ is some context parameter (can be omitted) such as home team advantage.

The parameters c and d are used to scale the ratings and are usually taken as c=10, d=400

Elo rating system

- Why are c=10, d=400.

- This is just for convention and scaling. If the Player A is 10 times more likely to beat Player B (Odds) then the difference of the Elo scores needs to be 400.
- Assume teams playing in neutral field $\theta = 0$

$$\gamma^H = \frac{1}{1+10^{(R^A-R^H)/400}}, \gamma^A = 1 - \frac{1}{1+10^{(R^A-R^H)/400}}$$

$$Odds(H\ beats\ A) = \frac{\gamma^H}{\gamma^A} = \frac{\frac{1}{1+10^{(R^A-R^H)/400}}}{\frac{10^{(R^A-R^H)/400}}{1+10^{(R^A-R^H)/400}}} = 10^{(R^H-R^A)/400}$$

if $R^H - R^A = 400$, then $Odds(H\ beats\ A)$=10

Elo rating system

- Several R packages for elo rating calculations
- We will use mostly “elo”

```
install.packages("PlayerRatings")
install.packages("elo")
install.packages("EloRating")
```

Elo rating system

- predict probability to win for player A and player B
- Calculate the odds to win

```
library(elo)
elo.prob(elo.A=1500, elo.B=1500)
```

```
## [1] 0.5
```

```
elo.prob(elo.A=1900, elo.B=1500)
```

```
## [1] 0.9090909
```

Elo rating system

$c = 10, d = 400$ parameters

```
elo.prob(elo.A=1900, elo.B=1500)
```

```
## [1] 0.9090909
```

```
# The odds
elo.prob(elo.A=1900, elo.B=1500)/(1-elo.prob(elo.A=1900, elo.B=1500))
```

```
## [1] 10
```

With the difference of 400 elo rating points, it is 10 times more likely that the Team A will win.

Elo rating system

The rating is updated with the following formula

$$R_t^H = R_{t-1}^H + K * (\alpha^H - \gamma^H)$$

where

$\boldsymbol{R_t^H}$ updated rating for the home team

$\boldsymbol{R_{t-1}^H}$ rating before the game

$\boldsymbol{\alpha^H}$ is the actual outcome for the game

$\boldsymbol{\gamma^H}$ is the predicted probability of Home team to win

$\boldsymbol{K}$ is a positive number

Elo rating system

The actual score for the home team α^H is given by

$$\alpha^H = \begin{cases} 1, & if\ home\ team\ won \\ 0.5, & if\ it\ was\ draw \\ 0, & Otherwise \end{cases}$$

In football/soccer the coefficient K can be updated based on the goal differential

$$K = k_0(1+\delta)^{\lambda}$$

Where δ is the absolute goal difference, k_0 and λ are positive constants

There are several suggestions for the value K, we will take it equal to 20

Elo rating system

example:

Tow players

- Player A, Elo rating of 1500
- Player B, Elo rating of 1600

$$R_t^H = R_{t-1}^H + K * \left(\alpha^H - \gamma^H\right)$$

```
# prob for A
elo.prob(elo.A=1500, elo.B=1600)
```

```
## [1] 0.359935
```

```
# For A
1500+20*(1-0.359935)
```

```
## [1] 1512.801
```

```
# For B
elo.prob(elo.A=1600, elo.B=1500)
```

```
## [1] 0.640065
```

```
1600+20*(0-0.640065)
```

```
## [1] 1587.199
```

Elo rating system

- The player A will get increase in rating by 12.8013
- The player B will get decrease in rating by 12.8013
- 12.8013 is calculated using function elo.update()

```
elo.calc(elo.A=1500, elo.B=1600, wins.A=1, k = 20)
```

```
##      elo.A    elo.B
## 1 1512.801 1587.199
```

```
elo.update(elo.A=1500,elo.B=1600,wins.A=1, k=20)
```

```
## [1] 12.8013
```

Elo rating system

Dataset on EPL games for seasons 2000-2018

```
library(SportsAnalytics270)
data(epl_00_18)
str(epl_00_18)
```

```
## 'data.frame':    6840 obs. of  6 variables:
##  $ Home  : chr  "Charlton Athletic" "Chelsea FC" "Coventry City" "Derby County" ...
##  $ Away  : chr  "Manchester City" "West Ham United" "Middlesbrough FC" "Southampton FC"
##  $ HG    : num  4 4 1 2 2 0 1 1 3 2 ...
##  $ AG    : num  0 2 3 2 0 0 0 0 1 0 ...
##  $ Round : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Season: num  2000 2000 2000 2000 2000 2000 2000 2000 2000 2000 ...
```

Elo rating system

score function from Elo package do a very simple calculation: Based on the game score returns a vector with the results, 1 if team A won, 0.5 if it was a draw and 0 if Team A lost

3 games, 2-0, 0-0, 0-1

```
score(2,0)
```

```
## [1] 1
```

```
score(0,0)
```

```
## [1] 0.5
```

```
score(0,1)
```

Elo rating system

Run Elo model

```
elos <- elo.run(score(HG, AG) ~ Home + Away,
        data = epl_00_18, k = 20)
elos_df <- as.data.frame(elos)
```

score(HG, AG) calculates the final score of the game based on Home Goal and Away Goal statistics

Elo rating system: Choosing K

- K values can be updated based on the importance of the games
- The Eloratings.net uses the following values for k for national teams

Competition	Multiplier
Friendly match	20
All minor tournaments	30
World Cup and continental qualifiers and major tournaments	40
Continental championship finals and major intercontinental tournaments	50
FIFA World Cup finals	60

Elo rating system

- Look at the dataframe
- initial values for elo is are set to 1500

```
head(elos_df)
```

```
##              team.A           team.B p.A wins.A update elo.A elo.B
## 1 Charlton Athletic  Manchester City 0.5    1.0     10  1510  1490
## 2        Chelsea FC  West Ham United 0.5    1.0     10  1510  1490
## 3     Coventry City Middlesbrough FC 0.5    0.0    -10  1490  1510
## 4      Derby County   Southampton FC 0.5    0.5      0  1500  1500
## 5      Leeds United       Everton FC 0.5    1.0     10  1510  1490
## 6    Leicester City      Aston Villa 0.5    0.5      0  1500  1500
```

Elo rating system

- The final elo ratings
- n=10 as there are 10 games each round

```
tail(elos_df, n=10)
```

```
##                 team.A                  team.B       p.A wins.A     update
## 6831        Burnley FC        AFC Bournemouth 0.5302023      0 -10.604045
## 6832    Crystal Palace   West Bromwich Albion 0.5687264      1   8.625472
## 6833 Huddersfield Town             Arsenal FC 0.2866354      0  -5.732709
## 6834      Liverpool FC Brighton & Hove Albion 0.7516706      1   4.966588
## 6835 Manchester United             Watford FC 0.7977292      1   4.045415
## 6836  Newcastle United             Chelsea FC 0.2200432      1  15.599136
## 6837    Southampton FC        Manchester City 0.1534775      0  -3.069549
## 6838      Swansea City             Stoke City 0.5145311      0 -10.290621
## 6839 Tottenham Hotspur         Leicester City 0.7497515      1   5.004970
## 6840   West Ham United             Everton FC 0.4239470      1  11.521060
##         elo.A    elo.B
## 6831 1536.584 1536.780
## 6832 1538.392 1473.081
## 6833 1482.521 1652.379
## 6834 1712.078 1509.745
## 6835 1723.775 1477.315
## 6836 1513.267 1701.894
## 6837 1514.255 1817.032
## 6838 1473.755 1484.236
## 6839 1743.655 1543.027
## 6840 1539.274 1569.492
```

Elo rating system

Get the final elos

```
elo_final <- final.elos(elos)
sort(elo_final, decreasing = T)
```

```
##         Manchester City       Tottenham Hotspur       Manchester United
##                1817.032                1743.655                1723.775
##           Liverpool FC             Chelsea FC             Arsenal FC
##                1712.078                1701.894                1652.379
##             Everton FC          Leicester City         West Ham United
##                1569.492                1543.027                1539.274
##         Crystal Palace        AFC Bournemouth             Burnley FC
##                1538.392                1536.780                1536.584
##         Southampton FC        Newcastle United  Brighton & Hove Albion
##                1514.255                1513.267                1509.745
##          Wigan Athletic             Stoke City         Birmingham City
##                1488.608                1484.236                1483.599
##       Huddersfield Town             Watford FC            Swansea City
##                1482.521                1477.315                1473.755
##    West Bromwich Albion          Blackpool FC        Bolton Wanderers
##                1473.081                1471.149                1467.713
##        Sheffield United          Ipswich Town       Charlton Athletic
##                1466.612                1463.073                1458.265
##               Fulham FC       Blackburn Rovers            Leeds United
##                1455.840                1447.924                1446.686
##               Hull City          Norwich City            Cardiff City
##                1441.610                1435.965                1435.650
##           Portsmouth FC         Coventry City             Reading FC
##                1431.484                1428.123                1423.624
##          Sunderland AFC       Middlesbrough FC Wolverhampton Wanderers
##                1418.103                1415.869                1409.884
##     Queens Park Rangers          Bradford City             Aston Villa
##                1405.656                1397.343                1370.075
##           Derby County
##                1294.607
```

Elo rating system

- Taking into account Home advantage
- θ Home advantage parameter = 200
- This means that for every game, 200 is added to home teams elo rating

$$\gamma^H = \frac{1}{1 + c^{(R^A - \theta - R^H)/d}}$$

$$\gamma^H = \frac{1}{1 + c^{(R^A - 200 - R^H)/d}}$$

Elo rating system

```
epl2 <- elo.run(score(HG, AG) ~ adjust(Home, 200) +
                                 Away,  data = epl_00_18, k = 20)
```

```
sort(final.elos(epl2), decreasing = T)
```

```
##         Manchester City       Tottenham Hotspur       Manchester United
##                1882.790                1799.793                1777.834
##             Chelsea FC           Liverpool FC             Arsenal FC
##                1762.026                1760.482                1700.735
##             Everton FC         Leicester City        West Ham United
##                1590.576                1557.335                1542.215
##         AFC Bournemouth             Burnley FC         Crystal Palace
##                1533.296                1531.229                1530.126
##          Southampton FC  Brighton & Hove Albion        Newcastle United
##                1528.952                1505.939                1500.823
##             Stoke City         Wigan Athletic        Birmingham City
```

Elo rating system

Defining dynamic K factor

Update the Elo rating formula using the following equation

$$R_t^H = R_{t-1}^H + K * G * (\alpha^H - \gamma^H)$$

Where G is a goal difference factor. Website Eloratings.net uses the following formula for G Factor

$$G = \begin{cases} 1, & N \leq 1 \\ 1.5, & N = 2 \\ \dfrac{N + 11}{8}, & N \geq 3 \end{cases}$$

Elo rating system

Adding the goal difference factor

Make goal difference factor

```
epl_00_18$G <- ifelse(abs(epl_00_18$HG-epl_00_18$AG)<=1, 1,
                      ifelse(abs(epl_00_18$HG-epl_00_18$AG)==2,1.5,
                             (abs(epl_00_18$HG-epl_00_18$AG)+11)/8))
```

```
head(epl_00_18,5)
```

```
##                 Home              Away HG AG Round Season     G
## 1 Charlton Athletic   Manchester City  4  0     1   2000 1.875
## 2        Chelsea FC   West Ham United  4  2     1   2000 1.500
## 3     Coventry City Middlesbrough FC  1  3     1   2000 1.500
## 4      Derby County    Southampton FC  2  2     1   2000 1.000
## 5      Leeds United        Everton FC  2  0     1   2000 1.500
```

Elo rating system

```
epl3<-elo.run(score(HG, AG) ~ adjust(Home, 200) + Away +
                              k(20*epl_00_18$G),  data = epl_00_18 )
```

```
sort(final.elos(epl3), decreasing = T)
```

```
##          Manchester City       Tottenham Hotspur       Manchester United
##                 1962.804                1865.159                1832.632
##            Liverpool FC             Chelsea FC             Arsenal FC
##                 1817.514                1779.042                1724.846
##              Everton FC          Leicester City          Crystal Palace
##                 1601.420                1570.200                1557.268
##              Burnley FC        West Ham United        AFC Bournemouth
##                 1549.243                1548.652                1530.760
##           Southampton FC       Newcastle United  Brighton & Hove Albion
##                 1523.939                1520.002                1506.210
##           Wigan Athletic             Stoke City             Watford FC
##                 1479.286                1476.806                1475.849
##            Swansea City      Huddersfield Town    West Bromwich Albion
##                 1468.772                1462.032                1461.984
##         Sheffield United          Ipswich Town         Birmingham City
##                 1455.443                1454.764                1454.399
##            Blackpool FC        Blackburn Rovers        Bolton Wanderers
##                 1452.081                1440.332                1436.024
```

Elo rating system

- Can we predict the actual outcome and not just the probability ?
- Result for the home team

$$Predicted\ outcome = \begin{cases} 1, & p \geq 0.55 \\ 0.5, & 0.45 < p < 0.55 \\ 0, & p \leq 0.45 \end{cases}$$

```
epl2$Predicted <- ifelse(epl2$p.A>=0.55, 1,
                         ifelse(epl2$p.A<=0.45,0,0.5))
library(caret)
confusion <- table(epl2$wins.A, epl2$Predicted)
```

```
sum(diag(confusion))/nrow(epl2)
```

```
## [1] 0.5124269
```

Elo rating system

- The teams change every season, players come and go
- The rating of the previous season cannot be carried fully to the new season
- The following formula is suggested by MARS rating system

New Season Starting Rating = 1500+(Previous Season Final Rating-1500)/2

For newly promoted teams

New Season Starting Rating = 1500+(Average Final Rating of relegated teams-1500)/2)

Elo rating system

To get elo rating working better

- Tune home advantage parameter
- Tune new teams initial elo's parameters
- different K parameters for different games
- Think of what else can influence the game

Elo rating system

Lot of different web-sites provides elo ratings in different sports

- Some mix of elo ratings and optimizations from FiveThirtyEight on NBA teams and games https://projects.fivethirtyeight.com/carmelo/
- Elo ratings from nfl https://projects.fivethirtyeight.com/complete-history-of-the-nfl/

- Football team elos
  - European clubs elo ratings http://clubelo.com/
  - for national teams http://eloratings.net